DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 13: The No-Arbitrage Principle

Arbitrage · The Law of One Price · A New Foundation for Pricing

September 7, 2026

Today's Agenda

1. A new way to price things

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2. Arbitrage: a precise definition

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3. The Law of One Price

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2. Arbitrage: a precise definition
3. The Law of One Price
4. Why this beats CAPM as a foundation

Learning Objectives

- Define arbitrage precisely; spot an arbitrage opportunity.

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- State and apply the Law of One Price.
- Explain why no-arbitrage pricing needs no risk-preference assumptions.
- Contrast this with CAPM's foundations.

Pricing Without Knowing Anyone's Feelings

The Umbrella and the Poncho-Plus-Hat

One approach: survey how much everyone hates getting wet, model utility, average.

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One approach: survey how much everyone hates getting wet, model utility, average. A different approach: an umbrella behaves just like a poncho + hat. If that combo costs Rs. 500, the umbrella must too – or everyone buys whichever's cheaper. No survey needed. No utility function needed.

Arbitrage Opportunity

Definition

A trading strategy that:

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A genuinely risk-free profit, from nothing.

Free Money on the Sidewalk

Why Arbitrage Can’t Persist

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Free Money on the Sidewalk

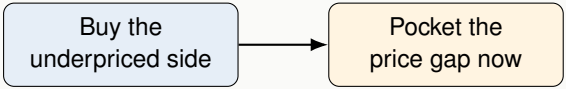
Why Arbitrage Can't Persist

A publicly visible risk-free profit doesn't last. Buying the cheap side pushes its price up; selling the expensive side pushes it down. The gap closes – often in seconds. "No free lunch on Wall Street" becomes a modeling assumption: price *as if* no arbitrage exists.

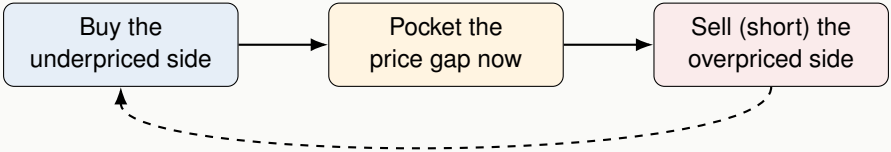
The Arbitrage Cycle

Buy the
underpriced side

The Arbitrage Cycle



The Arbitrage Cycle



The Law of One Price

Key Takeaway

Identical payoffs in **every** future state $\Rightarrow$ identical price today. Otherwise, arbitrage.

A Mispricing, and the Arbitrage It Creates

A, B both worth Rs. 110 in 1 year, certainly. A costs Rs. 100; B costs Rs. 102.

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Short-sell B	$+102$	-110

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Net	$+\mathbf{2}$	**0**

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Riskless Rs. 2 today, zero future exposure.

CAPM vs. No-Arbitrage Pricing

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Needs	Risk aversion, equilibrium, beliefs	Identical payoffs cost the same

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Argument	Equilibrium	Logical / constructive
Robustness	Fragile	Very few assumptions to challenge

The Payoff of This Lecture

Key Takeaway

Using *only* no-arbitrage, we will find an option's exact fair price – without the stock's expected return, without risk aversion, without equilibrium.

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- **Arbitrage**: risk-free profit from zero net investment.
- Competitive markets eliminate visible arbitrage almost instantly.
- **Law of One Price**: identical payoffs $\Rightarrow$ identical price.
- No-arbitrage pricing needs far fewer assumptions than CAPM.

Check Your Understanding

Discuss (2 minutes)

1. Two identical bonds paying Rs. 1000 in 1 year: X at Rs. 950, Y at Rs. 940. Build the arbitrage.

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3. Why must payoffs match in *every* state, not just on average?

Looking Ahead

Given a stock, a bond, and a derivative built from them...

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Replicating Portfolios – Lecture 14